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To:

Prof. Rock Jeng-Shing Chern / Editorial Office
Editor-in-Chief, Acta Astronautica
International Academy of Astronautics / Elsevier

**Subject: Submission of Original Research Article — "Project Setu: Paving the Road to 0.20c
Interstellar Flight"**

Dear Editor-in-Chief,

I am pleased to submit our original research manuscript titled "Project Setu: Paving the Road to 0.20c Interstellar Flight" for consideration as a regular research article in Acta Astronautica.

Achieving relativistic interstellar flight demands overcoming fundamental propulsion scaling limits. While beamed radiation propulsion offers a viable pathway to reach 0.20c without reaction mass penalties, macroscopic lightsails face severe coupled multi-physics hurdles—including extreme mechanical membrane stress under gigawatt-class laser thrust, severe localized thermal absorption leading to potential sublimation, and Doppler-shifted optical band mismatch during relativistic acceleration.

In this study, Project Setu establishes a comprehensive, high-fidelity 3D multi-physics numerical framework for a macroscopic 4.0-meter diameter relativistic lightsail accelerated by a 100 GW ground-based laser array ($\lambda_0 = 1064$ nm). The core contributions of this work include:

1. High-Fidelity Multi-Physics Coupling: Integration of 3D Maxwell FDTD wave electrodynamics, non-linear geometric membrane mechanics (Mindlin-Reissner SHELL181 elements), and Stefan-Boltzmann radiative thermal balance in ANSYS MAPDL and Ansys Lumerical.
2. Rigorous Grid Independence & ASME Standard Compliance: A four-level Richardson extrapolation and ASME GCI verification confirming a Grid Convergence Index of $GCI_{21} = 0.13\%$ and asymptotic convergence order $p = 1.95$, establishing numerical independence.
3. Structural & Thermal Survivability: Proof that maximum von Mises stress (530.88 MPa) maintains a 3.77x safety factor against the 2.0 GPa tensile yield limit of stoichiometric Si_3N_4 , while steady-state peak core temperature stabilizes at 923.02 K (providing a +1,247 K margin below sublimation).
4. Modal Stability: Determination of prestressed drumhead resonant frequencies (fundamental mode at 7.92 Hz) providing a 7.92x frequency buffer against sub-Hertz laser pointing jitter.
5. Real-World Spaceflight Telemetry Cross-Validation: Direct validation against operational in-flight telemetry from JAXA IKAROS (within 0.12%) and The Planetary Society / NASA LightSail 2 (within 2.13%), grounding the theoretical framework in empirical flight data.

This manuscript is original work, has not been published previously, and is not under consideration for publication elsewhere. All simulation scripts, ANSYS APDL macros, and datasets are fully documented for archival reproduction. Given Acta Astronautica's premier standing in advanced space propulsion, interstellar mission architectures, and spacecraft structural dynamics, we believe this manuscript will be of significant interest to your readership.

Thank you very much for your time, consideration, and handling of our manuscript.

Sincerely,

Prashant Suresh Kamble
Lead Researcher, Project Setu Initiative
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